

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

ME 19-205-0402 METROLOGY AND INSTRUMENTATION (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the concepts of measurements, limits, fits and tolerance.
 CO2: Learn about measurement techniques of angles, tapers and surface finish.
 CO3: Learn about applications of measuring instruments.
 CO4: Understand the measurement techniques for strain, force, torque, temperature, air pollution and nuclear radiation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A

(Answer ALL questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a) Explain why hole basis system is generally preferred.	3	L1	1	1,2	
	(b) Differentiate between workshop gauge and inspection gauge.	3	L2	1	1,2	
	(c) How is a sine bar specified?	3	L2	2	1,2	
	(d) Explain the principle of interferometry.	3	L1	2	1,2	
	(e) Differentiate between range and sensitivity.	3	L1	3	1,2	
	(f) With suitable examples explain the fidelity and the measuring lag.	3	L2	3	1,2	
	(g) What is a gauge factor? Explain its importance.	3	L2	4	1,2	
	(h) What is pyrometry? Briefly explain the theory of pyrometry.	3	L1	4	1,2	

PART B

(4 × 12 = 48)

- II. (a) Calculate the limits of tolerance and allowance for a 25 mm shaft and hole pair designated H8/d9. Given ;
 25 mm lies in the diameter steps 18 and 30 mm.
 For 'd' shaft the fundamental deviation = $-16 D^{0.44}$

Grades	IT5	IT6	IT7	IT8	IT9
Values	7i	10i	16i	25i	40i

- (b) Write a short note on limit gauges.
- OR**
- III. (a) Define unilateral and bilateral tolerances. Give examples for each.
 (b) Give a detailed classification of fits.
- OR**
- IV. (a) Explain the construction and use of a vernier and optical bevel protector.
 (b) What are the primary reasons for surface irregularities?
- OR**
- V. (a) Sketch and describe the optical system of the N.P.L. Flatness Interferometer.
 (b) Describe the principle and operation of Taylor-Hobson Talysurf surface roughness instrument.

(P.T.O.)

		Marks	BL	CO	PO
VI.	Do the modelling of a liquid in glass thermometer measurement system. What are the assumptions made during modeling? What is the order of the system?	12	L2	3	1,2
OR					
VII. (a)	Differentiate between linear and non-linear systems. Write down an equation describing a linear time invariant system and how it differs from a linear time variant system.	6	L2	3	1,2
(b)	The output of a system (y) is given by the following equation: $y = ax + b$ where x is the input and a and b are constants. Explain the system behavior (linear/non-linear) and explain why is it so?	6	L3	3	1,2
VIII. (a)	Describe a platform balance with a neat diagram.	6	L1	4	1,2
(b)	Explain the working principle of Pneumatic load cell with neat sketches.	6	L1	4	1,2
OR					
IX. (a)	Explain the working of a proving ring with a neat sketch.	6	L2	4	1,2
(b)	With the help of a neat diagram, explain the construction and working of an RTD.	6	L3	4	1,2

Blooms's Taxonomy Level's

L1 – 52.77 %, L2 – 45.98%, L3 – 1.25%.
